

Cheat-sheet PYTHON

Basic Commands

- `print("Hello, World!")` → Display output
 - `type(x)` → Get data type of `x`
 - `id(x)` → Get memory address of `x`
 - `help()` → Show documentation
 - `dir(obj)` → List object attributes and methods
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Variables & Data Types

- `int` → Whole numbers
 - `float` → Decimal numbers
 - `str` → Text values
 - `bool` → True or False values
 - `list` → Ordered, changeable collection
 - `tuple` → Ordered, unchangeable collection
 - `set` → Unordered, unique collection
 - `dict` → Key-value pair collection
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Built-in Functions

- `len(x)` → Length of `x`
- `max(x)` → Maximum value in `x`
- `min(x)` → Minimum value in `x`
- `sum(x)` → Sum of elements
- `sorted(x)` → Sorted list

- `range(start, stop, step)` → Generate sequence
 - `map(func, iterable)` → Apply function to elements
 - `filter(func, iterable)` → Filter elements
 - `zip(iter1, iter2)` → Combine iterables
 - `pop(i)` → Remove element at index `i`
 - `reverse()` → Reverse list order
 - `sort()` → Sort list
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Conditional Statements

- `for` → Iterates over a sequence
 - `while` → Runs while a condition is true
 - `break` → Exits the loop
 - `continue` → Skips to the next iteration
 - `pass` → Does nothing (placeholder)
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File Handling

- `open("file.txt", "r")` → Read file
- `open("file.txt", "w")` → Write file
- `open("file.txt", "a")` → Append file
- `file.read()` → Read file content
- `file.readline()` → Read one line
- `file.write("text")` → Write to file
- `file.close()` → Close file

Safe File Handling

```
withopen("file.txt","w")asf:  
f.write("Hello")
```

Functions

- `def` → Define a new function
- `return` → Return value from a function
- `lambda` → Create an anonymous function

Example:

```
defadd(a,b):  
returna+b
```

List Methods

- `append(x)` → Add element to end
- `insert(i, x)` → Insert at index `i`
- `remove(x)` → Remove first occurrence of `x`